LOYOLA UNIVERSITY

HIGHER TECHNICAL SCHOOL OF ENGINEERING

Robot Control Project

Author:
Pablo Carneado López

Professor:
Guillermo Bejarano Pellicer

Robot Programming and Control

November 2025

Degree in Robotics and Mechatronics

Contents

List of Figures

1. Introduction

This project aims to make an analysis of a given robot both, kinematically and dynamically, control it using different approaches and compare the results between them. The project will be done using the ”Robotics Toolbox” developed by Peter Corke in MATLAB.

2. Kinematic Analysis

2.1 Graphical representation

First of all it is needed to define the robot that will be worked on. Following the instructions of the assignment my configuration is D3. This combination of arm and wrist composes the following robot.

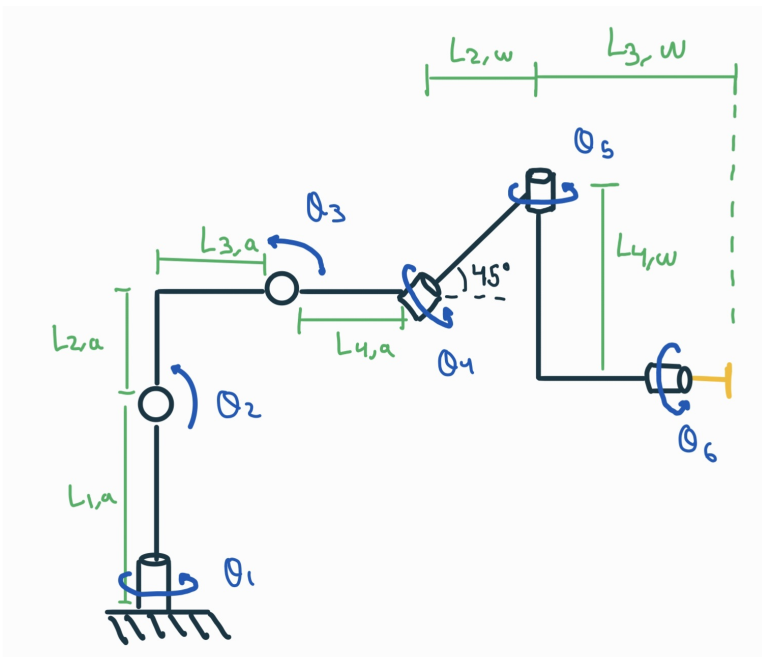


Figure 2.1: Representation of the robot of 6 DOF

2.2 Denavit-Hartenberg parameters

The Denavit-Hatermberg parameters are needed for all the computations and simulations of the robot. After following the steps of the algorithm the result is as follows.

	Joint	θ	d	a	α
r	1	θ_1	$L_{1,a}$	0	90°
r	2	$\theta_2+\beta+180^\circ$	0	$-\sqrt{L_2^2+L_3^2}$	0°
r	3	$\theta_3-(90-\beta+45)$	0	$-\frac{L_4}{\sqrt{2}}$	$+90^\circ$
r	4	θ_4+90	$\frac{L_{4a}}{\sqrt{2}}+L_{2,w}\cdot\sqrt{2}$	0	45°
r	5	θ_5	$-L_{4,w}$	0	270°
r	6	θ_6	$L_{3,w}$	0	0

$$\beta = \operatorname{atan}\left(\frac{L_{2a}}{L_{3a}}\right)$$

Figure 2.2: DH parameters of the 6 DOF robot

The Robot with all the axis identified is the following:

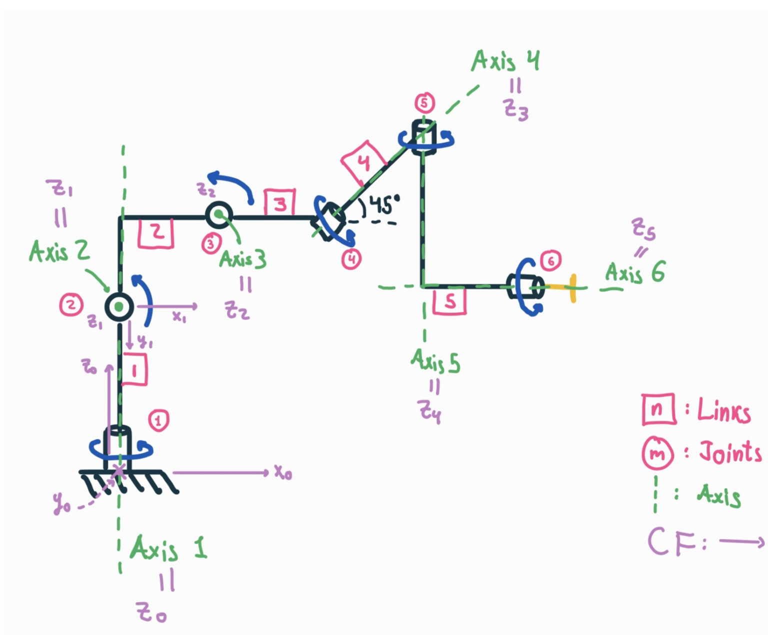


Figure 2.3: DH representation of the 6 DOF robot

The coordinate frames of the previous robot are as follows:

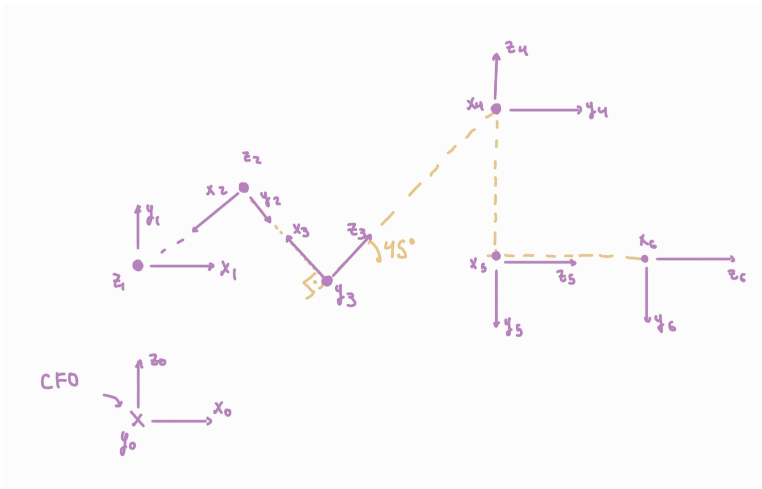


Figure 2.4: DH axis representation of the 6 DOF robot

2.3 Homogeneous transformation matrix

To translate the previous tables and drawings so that the computer can understand it's information, it is needed to compute the homogeneous transformation matrices (HTMs) of each adjacent joint pair.

$$^{0}A_1 = \begin{bmatrix} \cos(q1) & 0 & \sin(q1) & 0 \\ \sin(q1) & 0 & -\cos(q1) & 0 \\ 0 & 1 & 0 & L1a \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^{1}A_2 = \begin{bmatrix} -\cos(q2+\arctan(L2a/L3a)) & \sin(q2+\arctan(L2a/L3a)) & 0 & \cos(q2+\arctan(L2a/L3a))\sqrt{L2a^2+L3a^2} \\ -\sin(q2+\arctan(L2a/L3a)) & -\cos(q2+\arctan(L2a/L3a)) & 0 & \sin(q2+\arctan(L2a/L3a))\sqrt{L2a^2+L3a^2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^{2}A_3 = \begin{bmatrix} \cos(\frac{3\pi}{4} - q_3 + \arctan(\frac{L_{2a}}{L_{3a}})) & -\sin(\frac{3\pi}{4} - q_3 + \arctan(\frac{L_{2a}}{L_{3a}})) & 0 & -\frac{\sqrt{2}\,L_{4a}\cos(\frac{3\pi}{4} - q_3 + \arctan(\frac{L_{2a}}{L_{3a}}))}{2} \\ -\sin(\frac{3\pi}{4} - q_3 + \arctan(\frac{L_{2a}}{L_{3a}})) & -\cos(\frac{3\pi}{4} - q_3 + \arctan(\frac{L_{2a}}{L_{3a}})) & 0 & \frac{\sqrt{2}\,L_{4a}\sin(\frac{3\pi}{4} - q_3 + \arctan(\frac{L_{2a}}{L_{3a}}))}{2} \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^{3}A_4 = \begin{bmatrix} -\cos(q4) & \frac{\sqrt{2}\sin(q4)}{2} & -\frac{\sqrt{2}\sin(q4)}{2} & -\cos(q4)\left(\frac{\sqrt{2}\,L4a}{2} + \sqrt{2}\,L2w\right) \\ -\sin(q4) & -\frac{\sqrt{2}\cos(q4)}{2} & \frac{\sqrt{2}\cos(q4)}{2} & -\sin(q4)\left(\frac{\sqrt{2}\,L4a}{2} + \sqrt{2}\,L2w\right) \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^{3}A_4 = \begin{bmatrix} \cos(q_4+\frac{\pi}{2}) & -\frac{\sqrt{2}}{2}\sin(q_4+\frac{\pi}{2}) & \frac{\sqrt{2}}{2}\sin(q_4+\frac{\pi}{2}) & 0 \\ \sin(q_4+\frac{\pi}{2}) & \frac{\sqrt{2}}{2}\cos(q_4+\frac{\pi}{2}) & -\frac{\sqrt{2}}{2}\cos(q_4+\frac{\pi}{2}) & 0 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}\,L_{4a}}{2} + \sqrt{2}\,L_{2w} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^{4}A_{5} = \begin{bmatrix} \cos(q5) & 0 & -\sin(q5) & 0 \\ \sin(q5) & 0 & \cos(q5) & 0 \\ 0 & -1 & 0 & -L4w \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^{5}A_{6} = \begin{bmatrix} \cos(q6) & -\sin(q6) & 0 & 0 \\ \sin(q6) & \cos(q6) & 0 & 0 \\ 0 & 0 & 1 & L3w \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.4 Forward kinematics equations

For this part a simplification of the 6 DOF root will be used, a 3 DOF. This means using just the arm and not the wrist provided. The resultant robot is as follows:

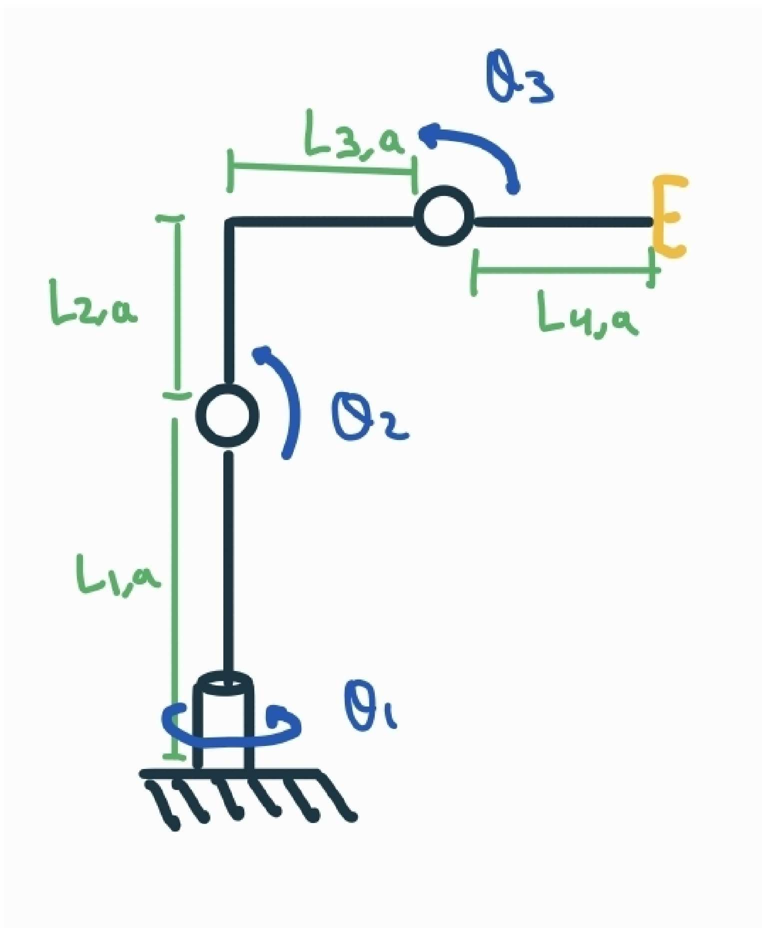


Figure 2.5: Representation of the 3 DOF robot

The axis of this robot are considerably simpler than the 6 DOF robot and are represented in the next figure:

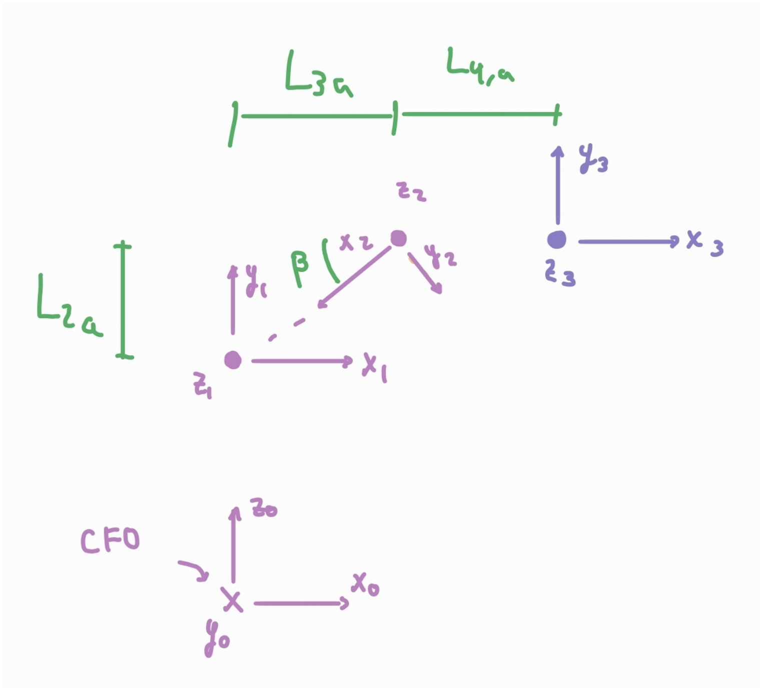
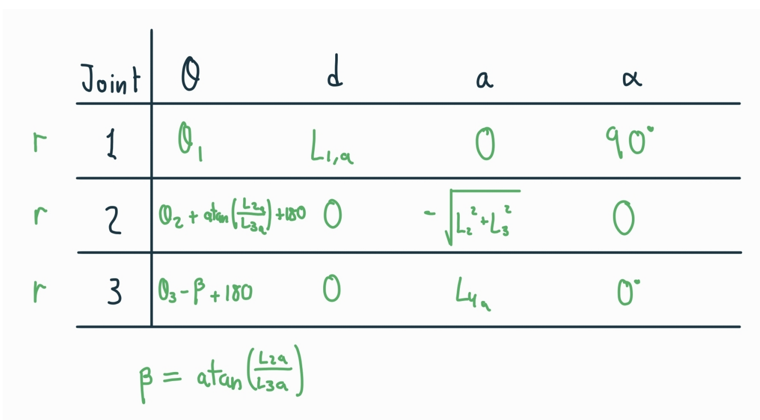


Figure 2.6: DH axis of the 3 DOF robot

This axis yield the following DH parameters for the 3 DOF robot:



	Joint	θ	d	a	α
r	1	θ_1	$L_{1,a}$	0	90°
r	2	$\theta_2 + \operatorname{atan}\left(\frac{L_{2a}}{L_{3a}}\right) + 180$	0	$-\sqrt{L_2^2 + L_3^2}$	0
r	3	$\theta_3 - \beta + 180$	0	L_{4a}	0°

$\beta = \operatorname{atan}\left(\frac{L_{2a}}{L_{3a}}\right)$

Figure 2.7: DH parameters of the 3 DOF robot

The equations for the forward kinematics are the translation vector from the HTM that goes from the joint 1 to 3.

$$\mathbf{X} = -\frac{\cos q_1 \left(2l_{3a}\cos q_2\sqrt{l_{2a}^2+l_{3a}^2} - 2l_{2a}\sin q_2\sqrt{l_{2a}^2+l_{3a}^2} + \sqrt{2}l_{2a}l_{4a}\cos(q_2-q_3) + \sqrt{2}l_{3a}l_{4a}\sin(q_2-q_3)\right)}{2l_{3a}\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}}$$

$$\mathbf{Y} = \frac{\sin(q_1)\left(l_{2a}^2 l_{4a}\cos(q_2+q_3) + l_{3a}^2 l_{4a}\cos(q_2+q_3) + l_{3a}^2\cos(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2} - l_{2a}l_{3a}\sin(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}\right)}{l_{2a}^2+l_{3a}^2}$$

$$\mathbf{Z} = \frac{l_{1a}l_{2a}^2 + l_{1a}l_{3a}^2 + l_{2a}^2 l_{4a}\sin(q_2+q_3) + l_{3a}^2 l_{4a}\sin(q_2+q_3) + l_{3a}^2\sin(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2} + l_{2a}l_{3a}\cos(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}}{l_{2a}^2+l_{3a}^2}$$

$$R_{checked} = \begin{bmatrix} \cos(q_2+q_3)\cos(q_1) & -\sin(q_2+q_3)\cos(q_1) & \sin(q_1) \\ \cos(q_2+q_3)\sin(q_1) & -\sin(q_2+q_3)\sin(q_1) & -\cos(q_1) \\ \sin(q_2+q_3) & \cos(q_2+q_3) & 0 \end{bmatrix}$$

To compute the Forward Kinematics for the 6 DOF robot it would be necessary to compute the HTM that goes from 1 to 6 and then from that matrix extract the information as it has been done for the 3 DOF case.

2.5 Linear velocity Jacobian and singularities

The linear velocity Jacobian for the first three joints of the robot can be obtained from the position vector of the end-effector

$$[J1] = \begin{bmatrix} -\frac{\sin(q_1)\left(l_{2a}^2 l_{4a}\cos(q_2+q_3)+l_{3a}^2 l_{4a}\cos(q_2+q_3)+l_{3a}^2\cos(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}-l_{2a}l_{3a}\sin(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}\right)}{l_{2a}^2+l_{3a}^2} \\ \frac{\cos(q_1)\left(l_{2a}^2 l_{4a}\cos(q_2+q_3)+l_{3a}^2 l_{4a}\cos(q_2+q_3)+l_{3a}^2\cos(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}-l_{2a}l_{3a}\sin(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}\right)}{l_{2a}^2+l_{3a}^2} \\ 0 \end{bmatrix}$$

$$[J2] = \begin{bmatrix} -\frac{\cos(q_1)\left(l_{2a}^2 l_{4a}\sin(q_2+q_3)+l_{3a}^2 l_{4a}\sin(q_2+q_3)+l_{3a}^2\sin(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}+l_{2a}l_{3a}\cos(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}\right)}{l_{2a}^2+l_{3a}^2} \\ -\frac{\sin(q_1)\left(l_{2a}^2 l_{4a}\sin(q_2+q_3)+l_{3a}^2 l_{4a}\sin(q_2+q_3)+l_{3a}^2\sin(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}+l_{2a}l_{3a}\cos(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}\right)}{l_{2a}^2+l_{3a}^2} \\ \frac{l_{2a}^2 l_{4a}\cos(q_2+q_3)+l_{3a}^2 l_{4a}\cos(q_2+q_3)+l_{3a}^2\cos(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}-l_{2a}l_{3a}\sin(q_2)\sqrt{\frac{l_{2a}^2+l_{3a}^2}{l_{3a}^2}}\sqrt{l_{2a}^2+l_{3a}^2}}{l_{2a}^2+l_{3a}^2} \end{bmatrix}$$

$$[J3] = \begin{bmatrix} -\frac{\cos(q_1)\left(l_{4a}\sin(q_2+q_3)l_{2a}^2+l_{4a}\sin(q_2+q_3)l_{3a}^2\right)}{l_{2a}^2+l_{3a}^2} \\ -\frac{\sin(q_1)\left(l_{4a}\sin(q_2+q_3)l_{2a}^2+l_{4a}\sin(q_2+q_3)l_{3a}^2\right)}{l_{2a}^2+l_{3a}^2} \\ \frac{l_{4a}\cos(q_2+q_3)l_{2a}^2+l_{4a}\cos(q_2+q_3)l_{3a}^2}{l_{2a}^2+l_{3a}^2} \end{bmatrix}$$

The Jacobian determinant of the 3-DOF manipulator becomes zero when the mechanism reaches configurations in which independent motion is lost. For the link lengths used in this project

$$L_1 = 0.54 \text{ m}, \quad L_2 = 0.50 \text{ m}, \quad L_3 = 0.45 \text{ m}, \quad L_4 = 0.40 \text{ m},$$

the singularities are obtained by imposing $\det(J) = 0$ on the analytical expression of the determinant.

The resulting singular configurations are:

- $\sin(q_3) = 0 \;\Rightarrow\; q_3 = k\pi, \quad k \in \mathbb{Z}.$
- $L_2\sin(q_2) = L_3\cos(q_2) \;\Rightarrow\; q_2 = \arctan\left(\frac{L_3}{L_2}\right).$

- $L_2 \sin(q_2 + q_3) = L_3 \cos(q_2 + q_3) \;\Rightarrow\; q_2 + q_3 = \arctan\left(\frac{L_3}{L_2}\right)$.

These conditions correspond to configurations in which joint axes become aligned, causing an instantaneous loss of one degree of freedom.

2.6 Joint trajectories representation

In order to further complicate the task and avoid singularities I have chosen the circle representation to be as intuitive as possible. Taking advantage of the rotational joint at the base I define the radius of the circle to be the length of both joints towards the end effector and the height the initial one as well. Specifically the properties are:

$$\text{radius} = L3_a + L4_a = 0.85\,\text{m}$$

$$\text{Centre position} = [\,0 \;\; 0 \;\; L1_a + L2_a\,]m$$

After applying this restrictions the circle ends up as follows:

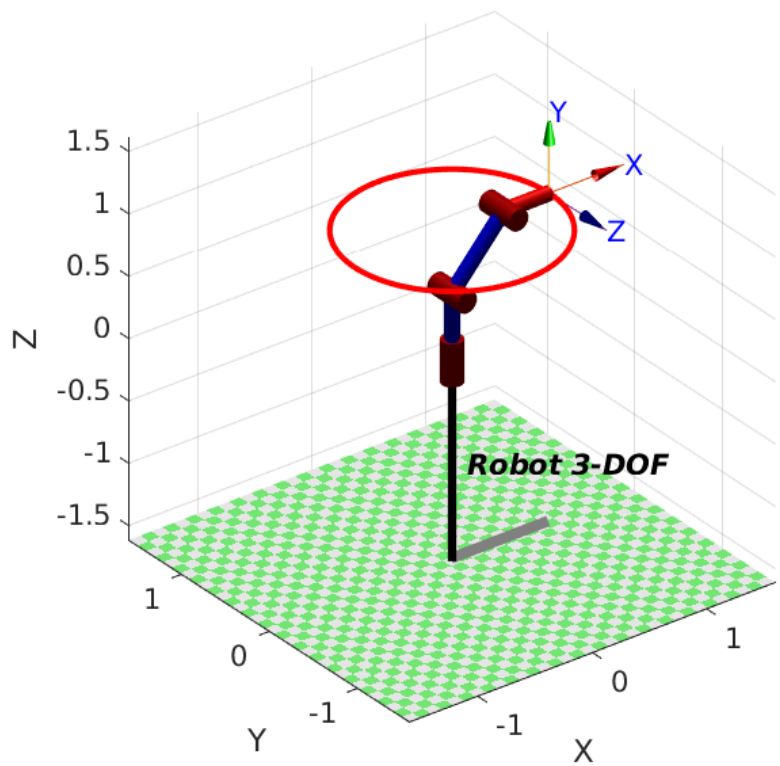


Figure 2.8: Circular Trajectory Representation

The joints position during this action is the following:

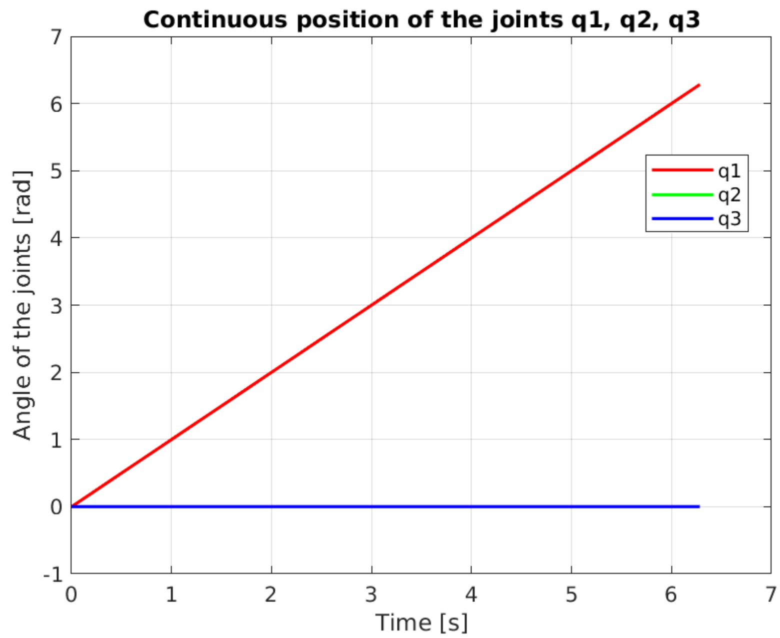


Figure 2.9: Joints Trajectory During Circle Representation

3. Dynamic Analysis

3.1 Computation of Dynamic Equations

For this computation I can use either the Lagrangian method od the Newton-Euler method, specifically for this case I use the Jacobian but either one can be applied.

This algorithm gives us the matrices needed to define the dynamic model of our robot. It is defined as follows:

$$\mathbf{Q} = \mathbf{M(q)}\,\ddot{\mathbf{q}} + \mathbf{C(q,\dot{q})} + \mathbf{G(q)}$$

So the algorithm gives as an output the matrices M,C and G. And the results obtained from them are the following:

$$\begin{bmatrix}M_1\end{bmatrix} = \begin{bmatrix} m_3\cos(q_1)^2(l_{4a}\cos(q_2+q_3)+l_{3a}\cos(q_2)-l_{2a}\sin(q_2))^2 + m_3\sin(q_1)^2(l_{4a}\cos(q_2+q_3)+ l_{3a}\cos(q_2)-l_{2a}\sin(q_2))^2 \\ +m_2\cos(q_2+\arctan\frac{l_{2a}}{l_{3a}})^2\cos(q_1)^2(l_{2a}^2+l_{3a}^2)+m_2\cos(q_2+\arctan\frac{l_{2a}}{l_{3a}})^2\sin(q_1)^2(l_{2a}^2+l_{3a}^2) \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix}M_2\end{bmatrix} = \begin{bmatrix} 0 \\ l_{2a}^2m_2+l_{2a}^2m_3+l_{3a}^2m_2+l_{3a}^2m_3+l_{4a}^2m_3+2l_{3a}l_{4a}m_3\cos(q_3)+2l_{2a}l_{4a}m_3\sin(q_3) \\ l_{4a}m_3(l_{4a}+l_{3a}\cos(q_3)+l_{2a}\sin(q_3)) \end{bmatrix}$$

$$\begin{bmatrix}M_3\end{bmatrix} = \begin{bmatrix} 0 \\ l_{4a}m_3(l_{4a}+l_{3a}\cos(q_3)+l_{2a}\sin(q_3)) \\ l_{4a}^2m_3 \end{bmatrix}$$

The Coriolis matrix ends up like follows:

$$\begin{bmatrix}C_1\end{bmatrix} = \begin{bmatrix} \frac{l_{2a}^2m_2\dot{q}_2\sin(2q_2)}{2}+\frac{l_{2a}^2m_3\dot{q}_2\sin(2q_2)}{2}-\frac{l_{3a}^2m_2\dot{q}_2\sin(2q_2)}{2}-\frac{l_{3a}^2m_3\dot{q}_2\sin(2q_2)}{2}-\frac{l_{4a}^2m_3\dot{q}_2\sin(2q_2+2q_3)}{2}-\frac{l_{4a}^2m_3\dot{q}_3\sin(2q_2+2q_3)}{2}-l_{2a}l_{4a}m_3\dot{q}_2\cos(2q_2+q_3)-\frac{l_{2a}l_{4a}m_3\dot{q}_3\cos(2q_2+q_3)}{2}-l_{3a}l_{4a}m_3\dot{q}_2\sin(2q_2+q_3)-\frac{l_{3a}l_{4a}m_3\dot{q}_3\sin(2q_2+q_3)}{2}-l_{2a}l_{3a}m_2\dot{q}_2\cos(2q_2)-l_{2a}l_{3a}m_3\dot{q}_2\cos(2q_2)+\frac{l_{2a}l_{4a}m_3\dot{q}_3\cos(q_3)}{2}-\frac{l_{3a}l_{4a}m_3\dot{q}_3\sin(q_3)}{2} \\ \frac{\dot{q}_1(l_{4a}^2m_3\sin(2q_2+2q_3)-l_{2a}^2m_2\sin(2q_2)-l_{2a}^2m_3\sin(2q_2)+l_{3a}^2m_2\sin(2q_2)+l_{3a}^2m_3\sin(2q_2)+2l_{2a}l_{4a}m_3\cos(2q_2+q_3)+2l_{3a}l_{4a}m_3\sin(2q_2+q_3)+2l_{2a}l_{3a}m_2\cos(2q_2)+2l_{2a}l_{3a}m_3\cos(2q_2))}{2} \\ \frac{l_{4a}m_3\dot{q}_1(l_{4a}\sin(2q_2+2q_3)-l_{2a}\cos(q_3)+l_{3a}\sin(q_3)+l_{2a}\cos(2q_2+q_3)+l_{3a}\sin(2q_2+q_3))}{2} \end{bmatrix}$$

$$\begin{bmatrix}C_2\end{bmatrix} = \begin{bmatrix} -\frac{\dot{q}_1}{2}\Big(l_{4a}^2m_3\sin(2q_2+2q_3)-l_{2a}^2m_2\sin(2q_2)-l_{2a}^2m_3\sin(2q_2)+l_{3a}^2m_2\sin(2q_2)+l_{3a}^2m_3\sin(2q_2)+2l_{2a}l_{4a}m_3\cos(2q_2+q_3)+2l_{3a}l_{4a}m_3\sin(2q_2+q_3)+2l_{2a}l_{3a}m_2\cos(2q_2)+2l_{2a}l_{3a}m_3\cos(2q_2)\Big) \\ l_{4a}m_3\dot{q}_3\big(l_{2a}\cos(q_3)-l_{3a}\sin(q_3)\big) \\ -l_{4a}m_3\dot{q}_2\big(l_{2a}\cos(q_3)-l_{3a}\sin(q_3)\big) \end{bmatrix}$$

$$\begin{bmatrix}C_3\end{bmatrix} = \begin{bmatrix} -\frac{l_{4a}m_3\dot{q}_1}{2}\Big(l_{4a}\sin(2q_2+2q_3)-l_{2a}\cos(q_3)+l_{3a}\sin(q_3)+l_{2a}\cos(2q_3+q_2)+l_{3a}\sin(2q_3+q_2)\Big) \\ l_{4a}m_3(\dot{q}_2+\dot{q}_3)\big(l_{2a}\cos(q_3)-l_{3a}\sin(q_3)\big) \\ 0 \end{bmatrix}$$

The last output that gives the algorithm is the gravity vector

$$G = \begin{bmatrix} 0 \\ gm_3(l_{4a}\cos(q_2+q_3)+l_{3a}\cos(q_2)-l_{2a}\sin(q_2))+gm_2\cos(q_2+\arctan\frac{l_{2a}}{l_{3a}})\sqrt{l_{2a}^2+l_{3a}^2} \\ gl_{4a}m_3\cos(q_2+q_3) \end{bmatrix}$$

3.2 Representation for Sinusoidal Torque Inputs

For this section it is necessary to define 3 sinusoidal curves, one for each joint of the robot, and observe the movement that it creates. For the amplitude of the wave I have selected for the first two joints an amplitude of 15 Nm and for the last one of 20 Nm. The offset of the wave is zero for all the joints but the second one because as it needs to carry the weight of the 2nd and 3rd link if it goes has negative values the second link will crash with the first one.

The torques applied are the following:

$$\tau_1(t) = 0 + 15\,\sin(t) = 15\,\sin(t)$$
$$\tau_2(t) = 20 + 15\,\sin(t)$$
$$\tau_3(t) = 0 + 20\,\sin(t) = 20\,\sin(t)$$

These inputs gives us the following movement:

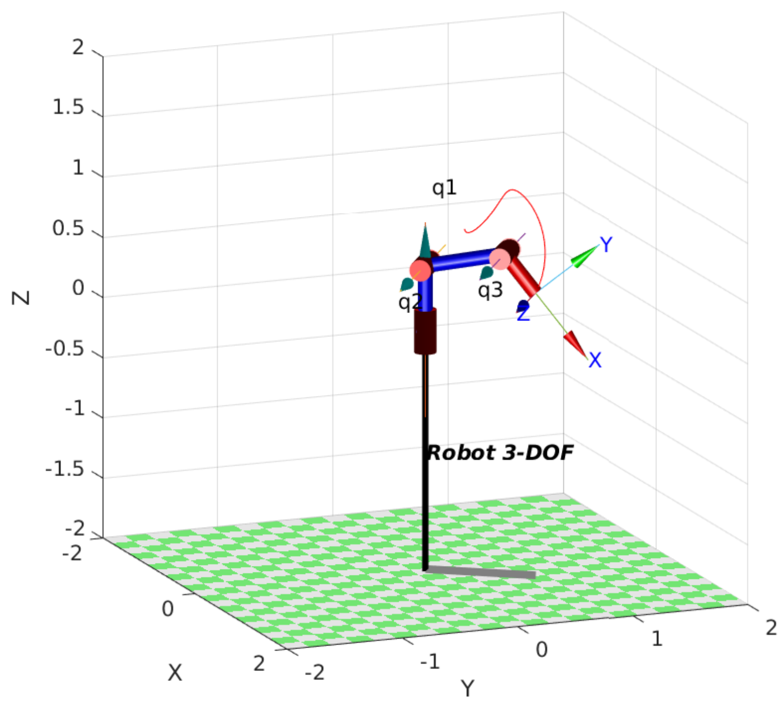


Figure 3.1: Sinusoidal Trajectory Representation

This movement has the following joints desplacement:

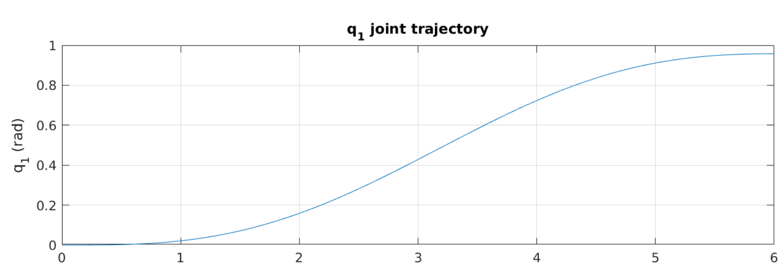


Figure 3.2: q1 Trajectory Representation

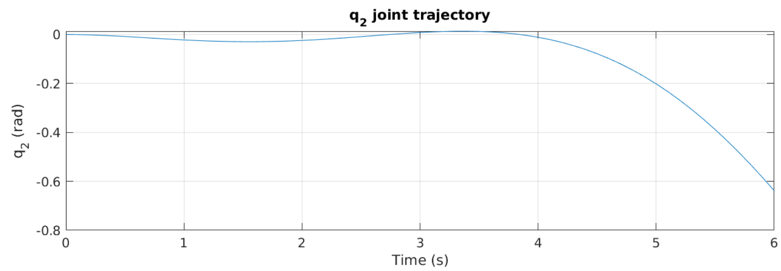


Figure 3.3: q2 Trajectory Representation

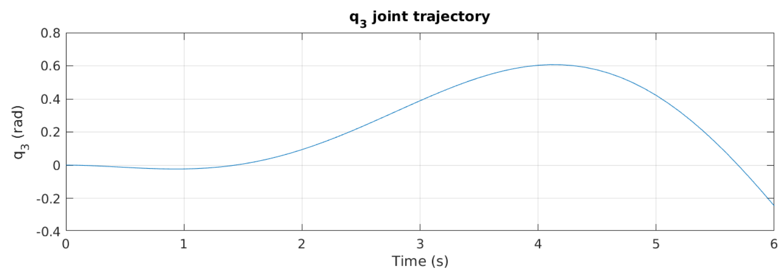


Figure 3.4: q3 Trajectory Representation

This movement is caused by this inputs:

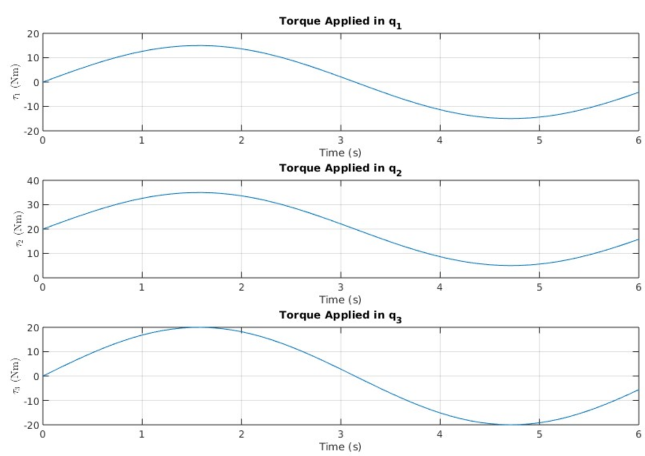


Figure 3.5: Sinusoidal Torques Representation

And the placement in the Cartesian coordinates is as follows:

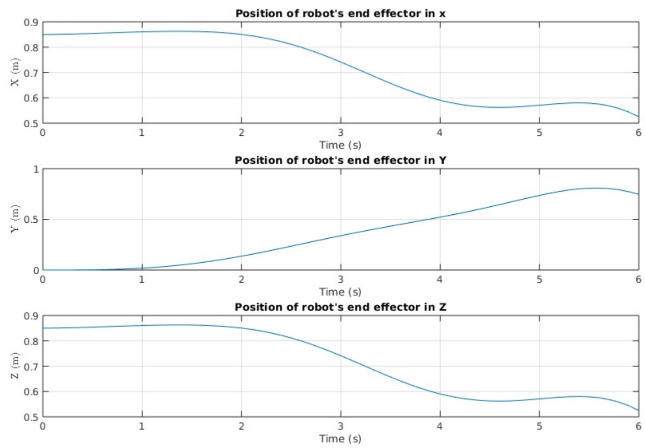


Figure 3.6: Cartesian position with sine inputs

4. Control

4.1 Independent Joint Control

4.1.1 Avoiding Singularities

In order to guarantee that the 3-DOF robot does not reach configurations arbitrarily close to its kinematic singularities, the reference trajectories for the joint controllers have been restricted to lie strictly inside a safe operating region. The singular configurations of the manipulator occur at

$$q_3 = k\pi, \qquad q_2 = \arctan\left(\frac{l_2}{l_3}\right), \qquad q_2 + q_3 = \arctan\left(\frac{l_2}{l_3}\right),$$

where approaching any of these conditions leads to a loss of rank in the Jacobian matrix and thus to reduced mobility or unbounded joint velocities.

To avoid operating near these configurations, the joint references are constrained to the intervals

$$q_1 \in [0,\ \tfrac{\pi}{2}], \qquad q_2 \in [0,\ \tfrac{\pi}{8}], \qquad q_3 \in [0,\ \tfrac{\pi}{8}].$$

By selecting reference values within these bounded regions, the commanded joint trajectories remain sufficiently far from the singularity loci, ensuring well-conditioned kinematics and avoiding numerical or control issues associated with singularities.

4.1.2 Sample Time Selection

The selection of an appropriate sample time (T_s) is fundamental for the digital implementation of the controllers. An inadequate sample time can lead to instability, poor tracking performance, or aliasing effects.

Theoretical Foundation

Each joint i of the 3-DOF robot can be modelled as a first-order system characterized by a time constant:

$$\tau_i = \frac{J_{m,i}}{B_{m,i}} \tag{4.1}$$

where $J_{m,i}$ is the motor inertia and $B_{m,i}$ is the viscous friction coefficient.

Calculation Methods

Three complementary methods were used to determine the sample time:

Method 1: Time Constant Criterion

The sample time should be at least 10 times faster than the fastest system dynamics:

$$T_{s,\tau} = \frac{1}{10} \min(\tau_1, \tau_2, \tau_3) \tag{4.2}$$

Method 2: Bandwidth Criterion
Based on the Nyquist-Shannon sampling theorem with a safety factor of 10:

$$T_{s,\omega} = \frac{2\pi}{10 \cdot \max(\omega_{b,i})} \tag{4.3}$$

Method 3: Settling Time Criterion
The settling time (2% criterion) is $t_{s,i} = 4\tau_i$. Sampling at least 100 times during settling:

$$T_{s,t_s} = \frac{4}{100} \min(\tau_i) \tag{4.4}$$

After computing all the possible sampling times the smallest one is selected and used to calculate the model.

4.1.3 Decentralized PD

The PD controller has two parameters that must be tuned to work properly. These parameters have been chosen so that the steady-state error for a unit ramp input is lower than 1%. The values for the proportional and derivative gains are:

$$K_d = \begin{bmatrix} 107.1429 & 131.5789 & 36.3636 \end{bmatrix} \frac{\mathrm{V} \cdot \mathrm{s}}{\mathrm{rad}}, \quad K_p = \begin{bmatrix} 0.1400 & 0.1900 & 0.2200 \end{bmatrix} \frac{\mathrm{V}}{\mathrm{rad}}$$

These parameters correspond to the ideal form of the PD controller, where the control law is

$$u(t) = K_p\, e(t) + K_d\, \frac{de(t)}{dt}$$

with $u(t)$ in volts $[V]$ and $e(t)$ in radians $[rad]$.

This parameters controlling the system gives this results:

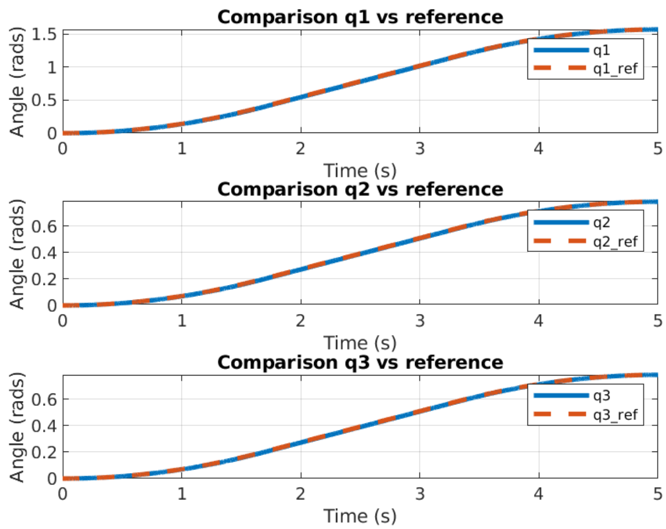


Figure 4.1: PD reference following

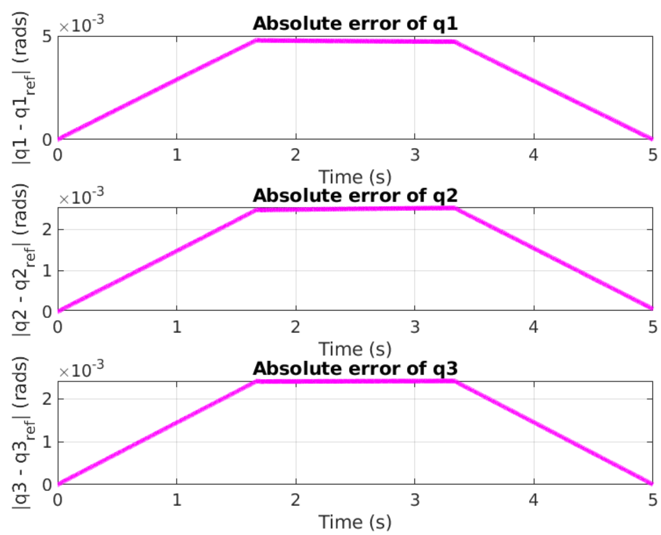


Figure 4.2: PD error

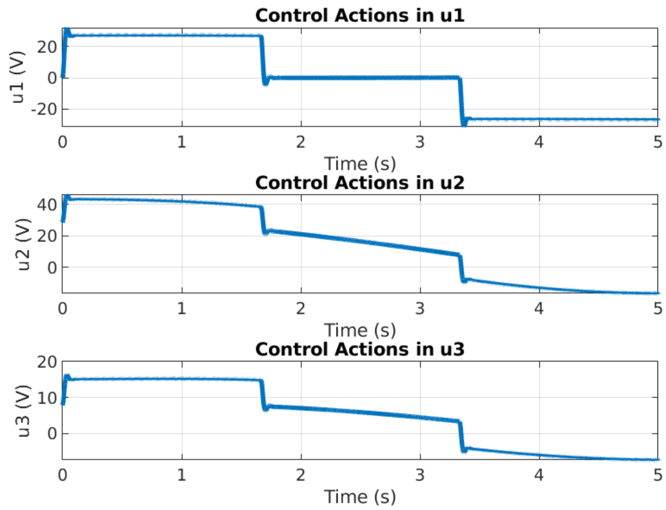


Figure 4.3: PD control actions

To prove the robustness of the controller I will add an uncertainty to the robot of the 20 percent that makes the simulated robot differ from the data that has been used to obtain the controllers. The results obtained after applying this uncertainty are the following:

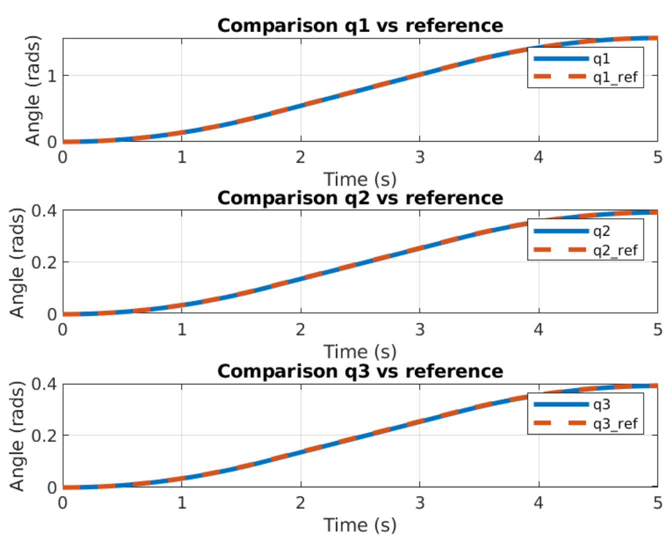


Figure 4.4: PD reference following with uncertainties

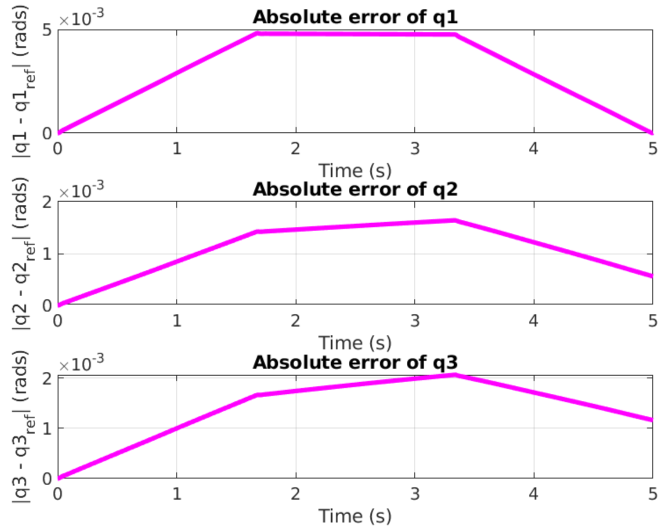


Figure 4.5: PD error with uncertainties

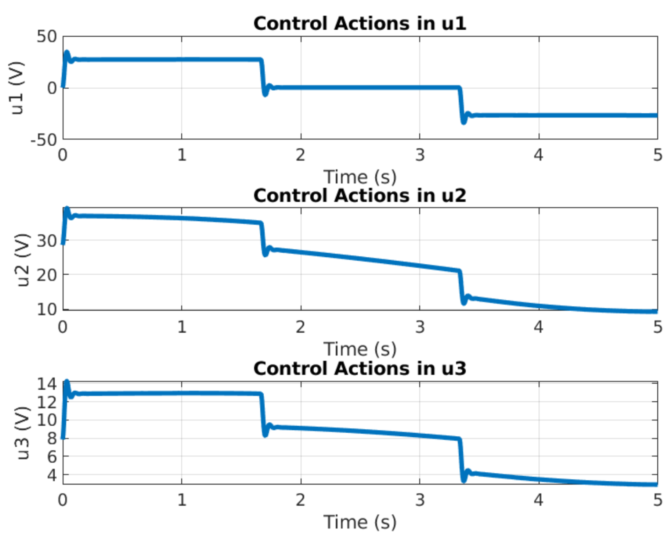


Figure 4.6: PD control actions with uncertainties

As it can be seen in the graphs the performance of the controller is quite good, from the information on the first figures, the reference following ones, the difference between the reference and the actual path is barely noticeable, once we have a look at the absolute error graph is when it can be seen that the error is higher in the middle of the trajectory. The value of the error is almost negligible in both cases; with and without disturbances but in the perturbed case it can be seen that the error is not reduced a much as in the normal case.

The maximum error that the system has is of 5 to the power of minus 3, as we are talking about radians that means less than 0.3 degrees, it wouldn't be noticeable for a normal human at a distance of 1 meter.

The control actions show some overshoot but that shouldn't be a problem with normal DC motors as they can handle changes in the input voltages fairly good, and the rest of the control action is very plain so this does not consider a problem for the life of the parts of the robot.

4.1.4 Decentralized PID

To achieve zero steady-state error, a PID controller has been implemented in its ideal form:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \frac{de(t)}{dt}$$

where $e(t)$ represents the position error of each joint in radians, and $u(t)$ is the control signal in volts applied to the motors.

- **Proportional term:** $u_P = K_p e(t)$
  To ensure the output is in volts, K_p must have units [V/rad].
- **Integral term:** $u_I = K_i \int_0^t e(\tau)\, d\tau$
  The integral of the error signal has units $[rad \cdot s]$, so K_i must have units [V/(rad·s)] to keep the output in volts.
- **Derivative term:** $u_D = K_d \frac{de(t)}{dt}$
  The derivative of the error has units $[rad/s]$, so K_d must have units [V·s/rad] to maintain consistency in the output units.

The design of the PID has been done so that the

$$\Delta M_f < 90$$

for a selected frequency. The phase margins of the controlled system of each joint at that frequency has to be higher than 80 this can be seen with the margin command in matlab as follows:

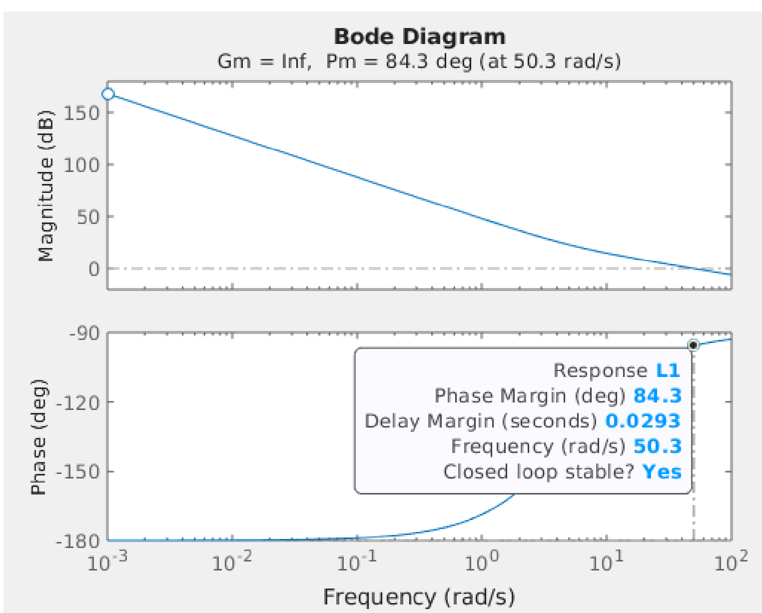


Figure 4.7: PID reference following

The obtained parameters, expressed in the corresponding units and for the ideal form of the PID controller, are:

$$P = \begin{bmatrix} 3.7570 \\ 6.2595 \\ 2.0110 \end{bmatrix} \frac{\text{V}}{\text{rad}}, \quad I = \begin{bmatrix} 0.0093 \\ 0.0076 \\ 0.0273 \end{bmatrix} \frac{\text{V}}{\text{rad·s}}, \quad D = \begin{bmatrix} 0.1996 \\ 0.1997 \\ 0.1989 \end{bmatrix} \frac{\text{V·s}}{\text{rad}}$$

This parameters make the system behave this way:

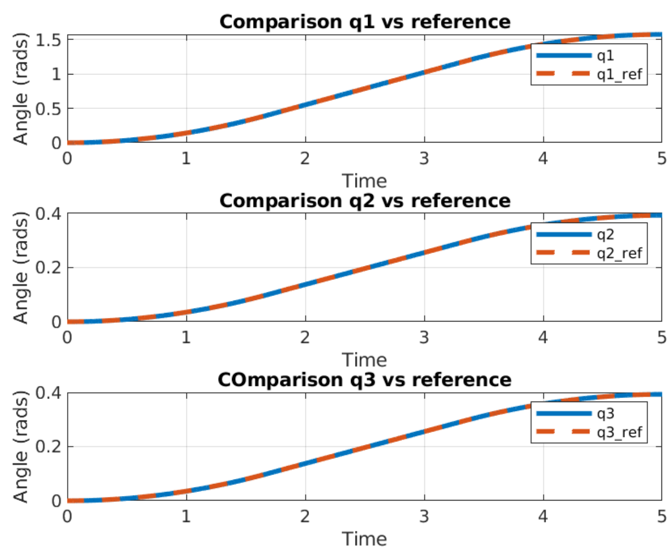


Figure 4.8: PID reference following

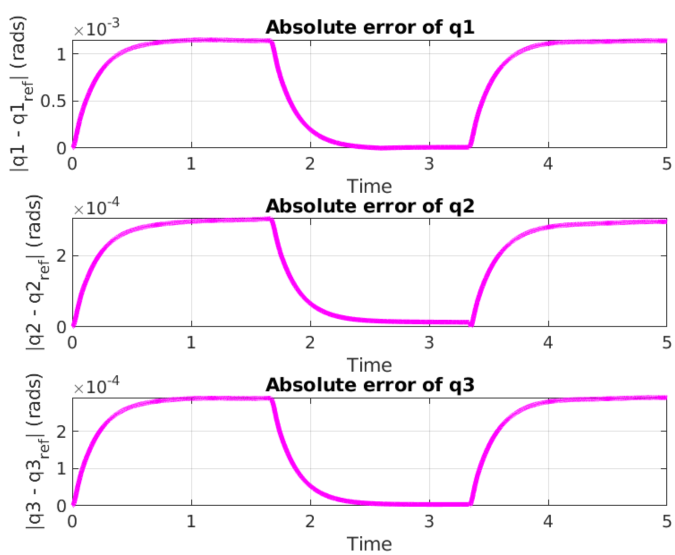


Figure 4.9: PID error

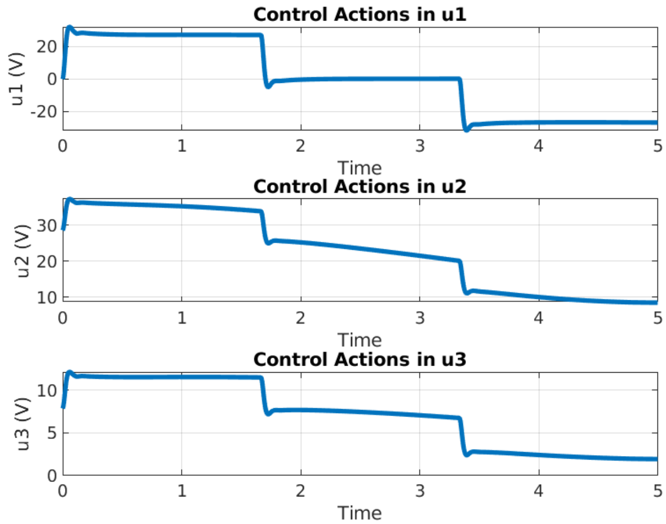


Figure 4.10: PID control actions

To prove the robustness of the controller I will add an uncertainty to the robot of the 20 percent that makes the simulated robot differ from the data that has been used to obtain the controllers. The results obtained after applying this uncertainty are the following:

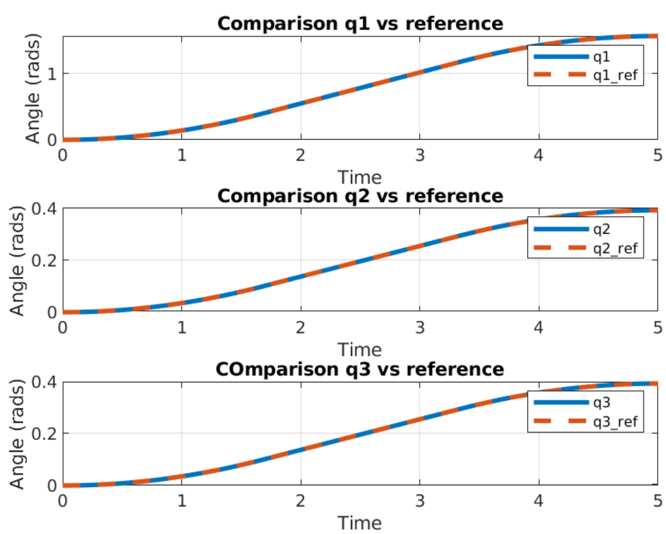


Figure 4.11: PID reference following with uncertainties

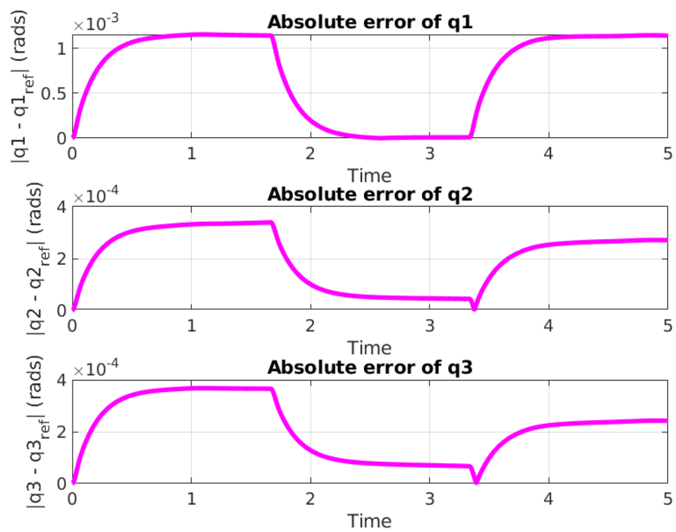


Figure 4.12: PID error with uncertainties

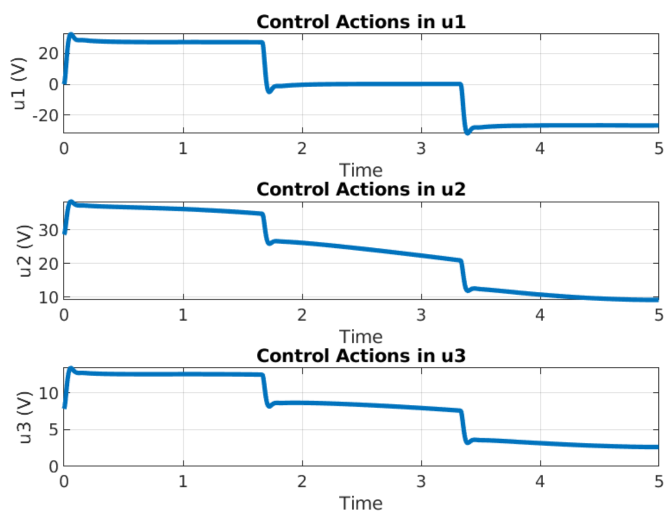


Figure 4.13: PID control actions with uncertainties

As it can be observed in the figures, the performance of the PID controller is very good. From the reference–following plots, the difference between the reference trajectory and the actual joint motion is still barely noticeable. However, when analysing the absolute error, a clear difference with respect to the PD controller appears: due to the integral action, the error no longer presents a trapezoidal shape but instead follows a first–order decay profile, decreasing smoothly along the trajectory.

The magnitude of the tracking error remains very small in both scenarios, with and without disturbances. In the perturbed case, the disturbances introduce a slight overshoot in a portion of the error curve, although the integral term compensates for the steady–state component effectively, preventing the error from growing significantly compared to the nominal behaviour.

The maximum error is of the same order of magnitude as before, around 10^{-3} radians, which corresponds to well below a degree of deviation. Such a small angular difference would not be noticeable to an observer at a typical operating distance, confirming that the controller preserves high precision even under external perturbations.

Regarding the control action, the overshoot observed in the PD case has been reduced thanks to the incorporation and tuning of the integral term. The resulting control effort is smoother, with smaller peaks and a more regular temporal evolution. This behaviour is favourable for standard DC motors, since the reduced overshoot lowers mechanical stress and contributes positively to the long–term reliability of the robotic system.

4.1.5 Independent Nested Control

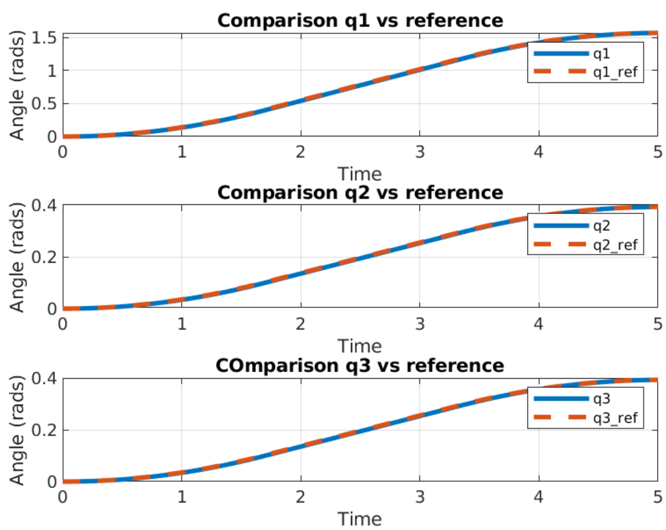


Figure 4.14: nested reference following

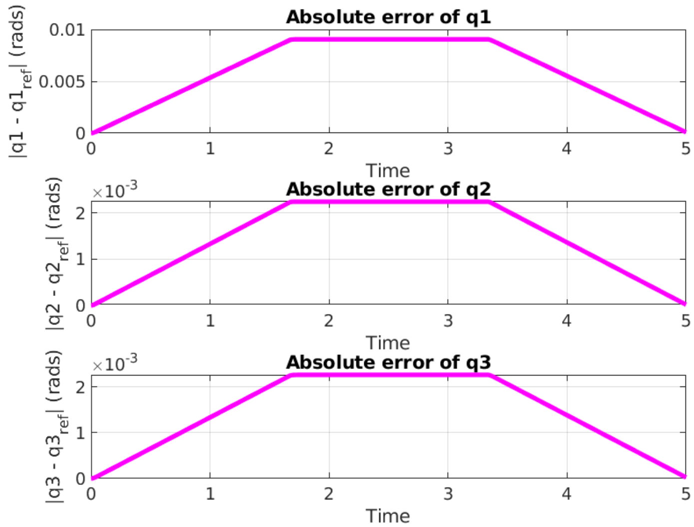


Figure 4.15: nested error

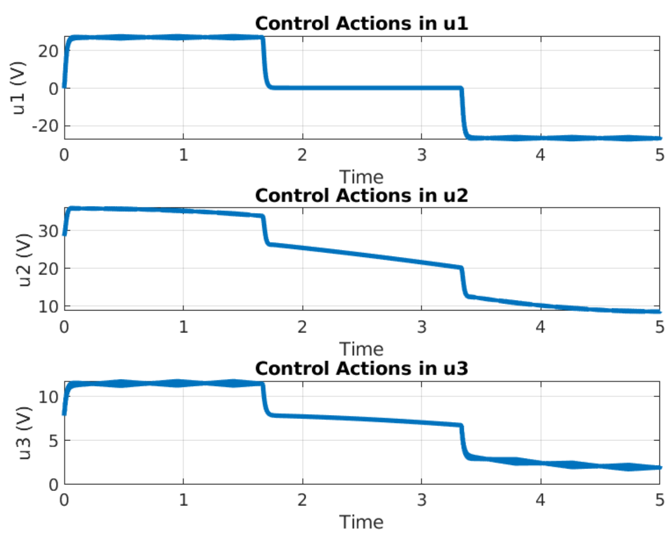


Figure 4.16: nested control actions

To prove the robustness of the controller I will add an uncertainty to the robot of the 20 percent that makes the simulated robot differ from the data that has been used to obtain the controllers. The results obtained after applying this uncertainty are the following:

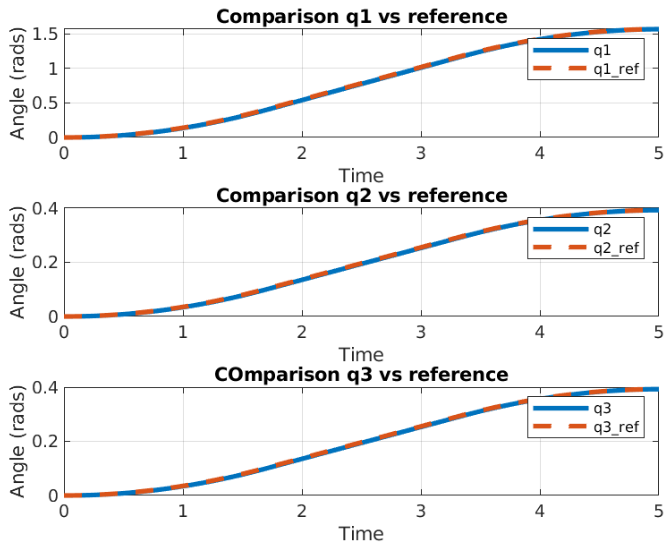


Figure 4.17: nested reference following with uncertainties

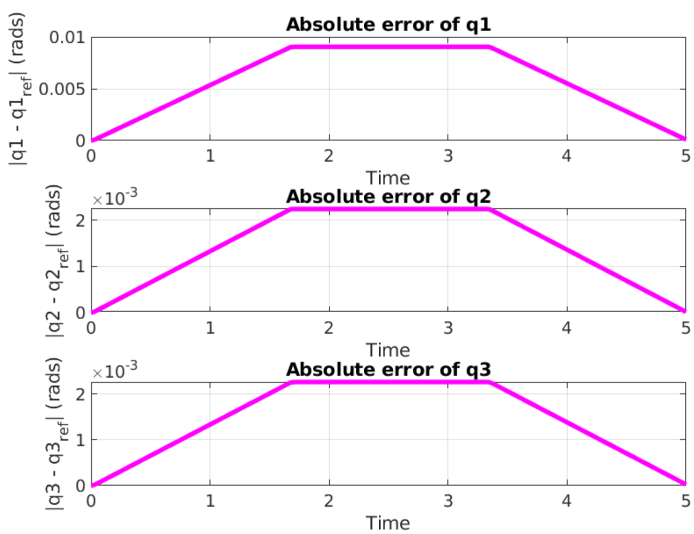


Figure 4.18: nested error with uncertainties

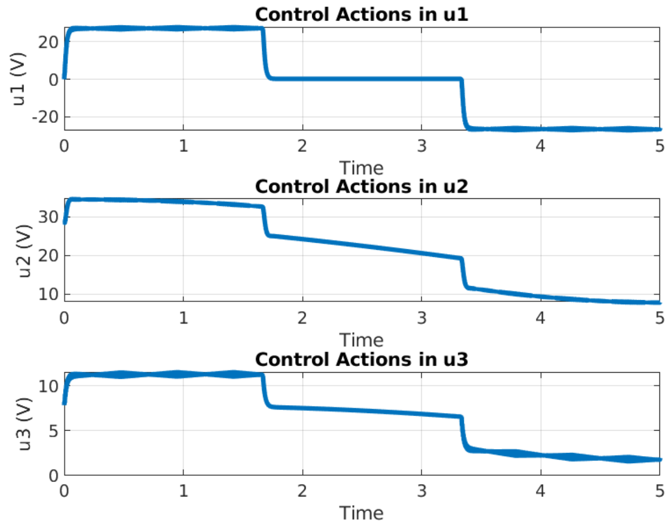


Figure 4.19: nested control actions with uncertainties

As it can be seen in the figures, the performance of the nested control scheme is comparable to that obtained with the PD and PID controllers. The reference–following behaviour is accurate, and the joint trajectories closely match their corresponding references throughout the entire motion.

When analysing the absolute error, the shape resembles more closely what was observed in the PD case, showing a trapezoidal profile rather than the first–order decay introduced by the integral action in the PID design. In general, the errors remain small; however, in the presence of uncertainties, the first joint exhibits a slightly higher deviation. The increase is not drastic, around 0.01 radians (a bit more than half a degree), which is still a reasonable value. This behaviour is expected, since the first joint must cover a significantly larger range of motion compared to the other two, and unlike them, it is not directly constrained by singularity–avoidance limits. Therefore, it is natural that this joint accumulates more sensitivity to disturbances.

The overall precision of the controller remains satisfactory, and the system continues to behave reliably even under perturbed conditions.

Regarding the control action, the overshoot observed in the PD and PID cases has been practically eliminated. The nested structure provides smoother input signals, with no significant peaks and a more uniform temporal evolution. This is beneficial for the actuators, reducing mechanical stress and contributing to the long–term health of the system components.

4.2 Model-Based Control

4.2.1 Feed-forward Control

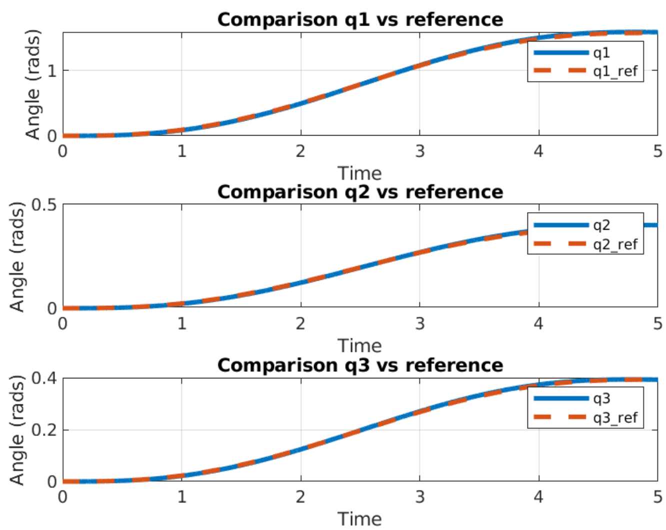


Figure 4.20: ff reference following

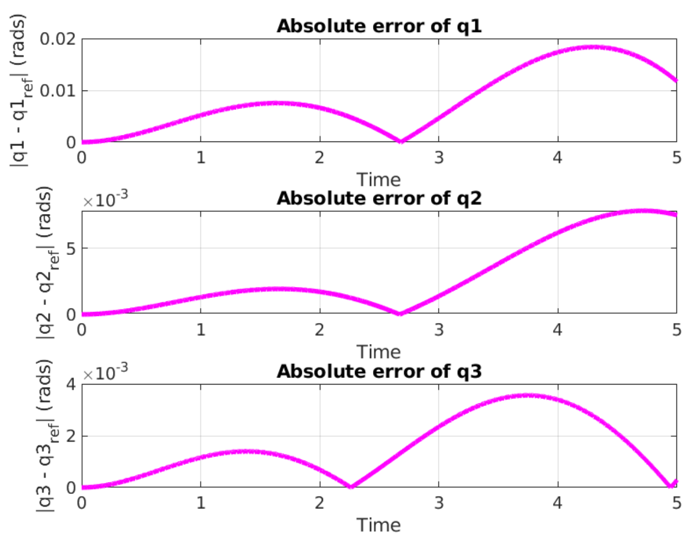


Figure 4.21: ff error

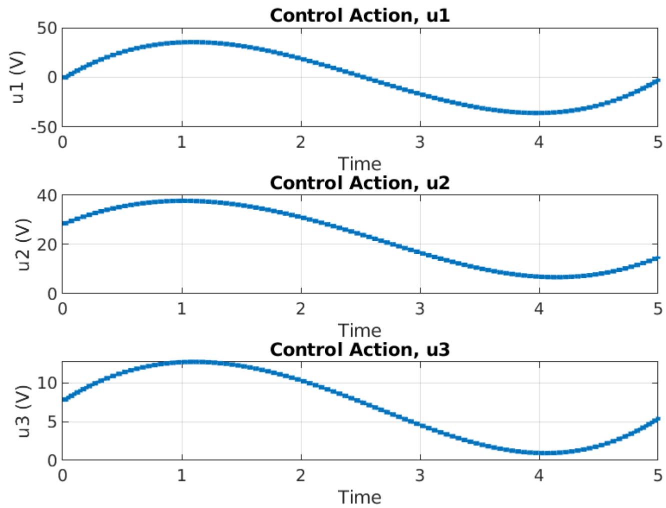


Figure 4.22: ff control actions

To prove the robustness of the controller I will add an uncertainty to the robot of the 20 percent that makes the simulated robot differ from the data that has been used to obtain the controllers. The results obtained after applying this uncertainty are the following:

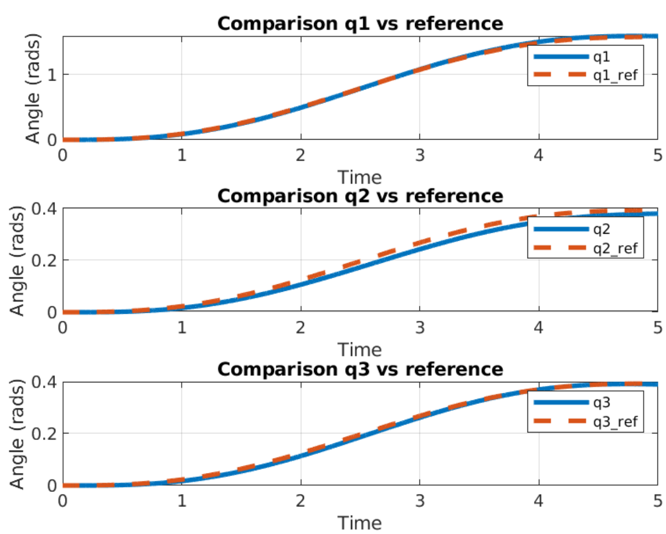


Figure 4.23: ff reference following with uncertainties

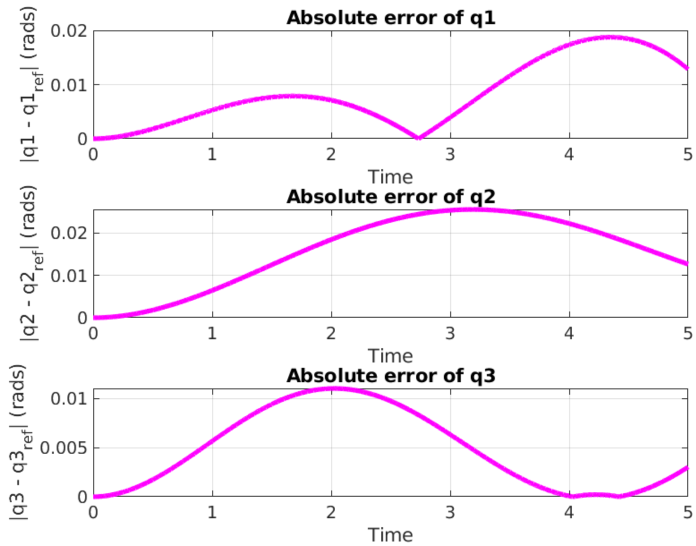


Figure 4.24: ff error with uncertainties

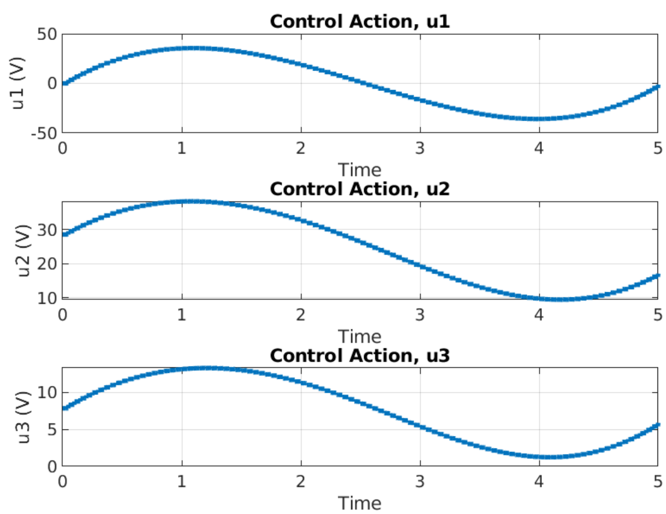


Figure 4.25: ff control actions with uncertainties

The feed-forward model-based controller exhibits a noticeably higher tracking error compared to the previously analysed PD, PID and nested controllers. In particular, the maximum deviation occurs in the first joint and reaches approximately 0.02 radians (about 1.15°). Although this error is not catastrophic, it indicates that the pure feed-forward strategy is not suitable for high-precision tasks where sub-degree accuracy is required. The remaining joints show smaller errors on the order of 4×10^{-3} to 5×10^{-3} radians, which are comparable to the errors observed with the other controllers.

The control signals produced by the feed-forward controller are substantially larger and notably smoother than those of the feedback-only controllers. Their temporal profiles are close to sinusoidal and present amplitudes on the order of several tens (40 in the plotted units). This behaviour is consistent with a controller that directly commands the motor input to track the nominal model torque/voltage required by the desired trajectory.

A plausible explanation for both the increased control amplitudes and the degraded precision is the absence of gearbox reduction in the modelling or actuation chain: because the feed-forward law is applied directly between motors and links (without the mechanical advantage of gearboxes), the motors must supply larger inputs to produce the same joint motion. Consequently, the system becomes more sensitive to model inaccuracies and external disturbances.

Indeed, disturbances are more evident in the feed-forward case, and their effect can be observed already in the reference-following plots. This highlights a fundamental limitation of pure feed-forward control: without adequate feedback compensation, unmodelled dynamics, friction mismatches and external perturbations translate directly into tracking errors. Therefore, while the feed-forward controller can provide smooth and large control efforts that follow the nominal dynamics, it should be complemented with feedback action when precision and disturbance rejection are required.

4.2.2 Computed Torque

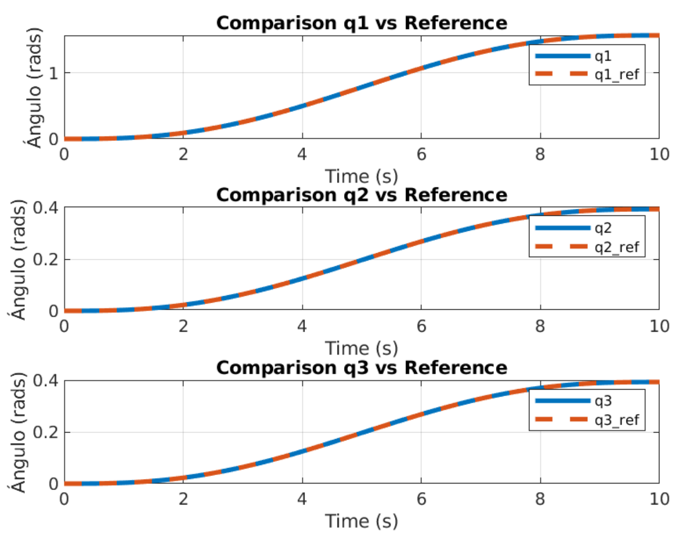


Figure 4.26: ct reference following

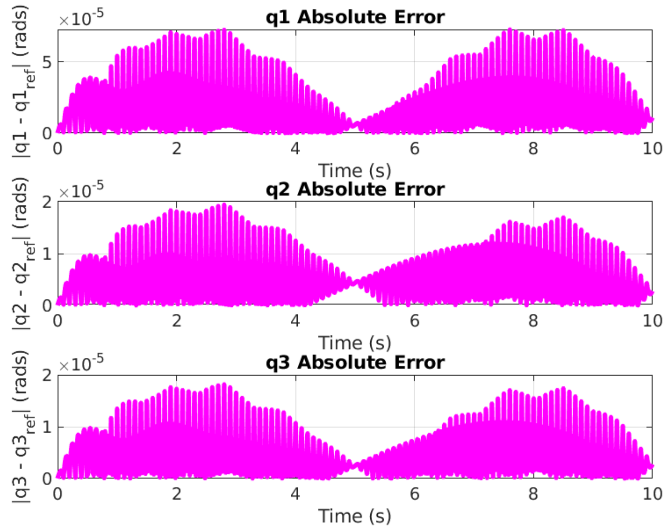


Figure 4.27: ct error

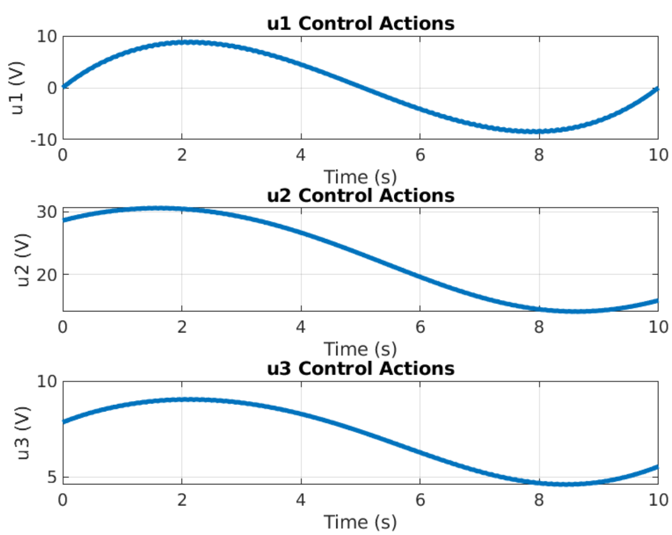


Figure 4.28: ct control actions

To prove the robustness of the controller I will add an uncertainty to the robot of the 20 percent that makes the simulated robot differ from the data that has been used to obtain the controllers. The results obtained after applying this uncertainty are the following:

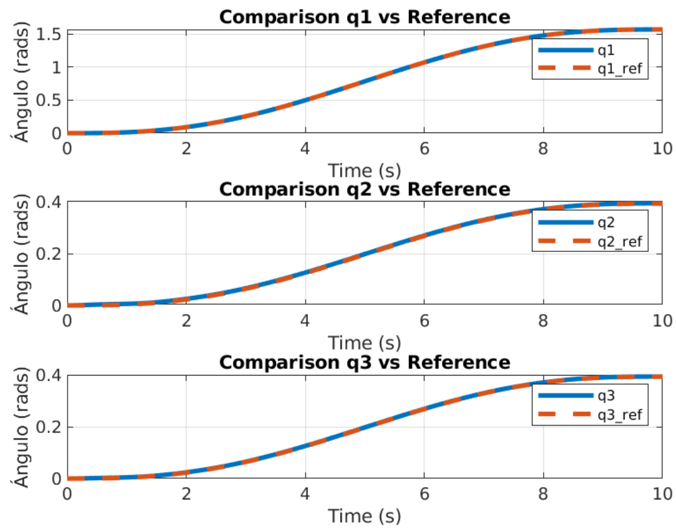


Figure 4.29: ct reference following with uncertainties

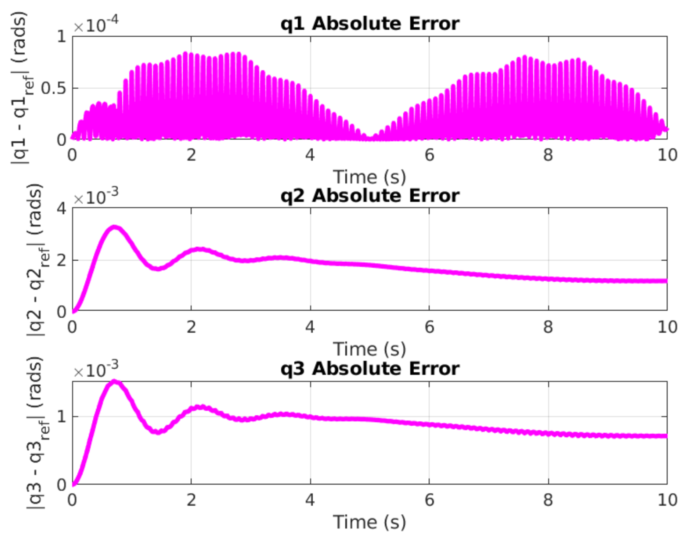


Figure 4.30: ct error with uncertainties

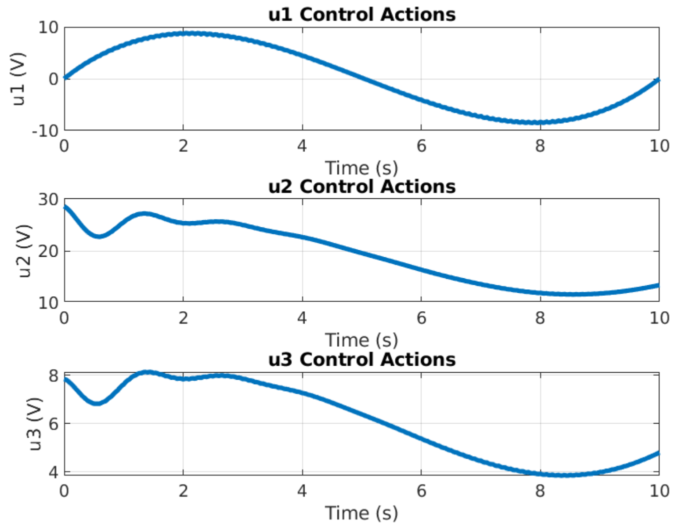


Figure 4.31: ct control actions with uncertainties

The computed-torque controller exhibits excellent nominal accuracy while showing some high-frequency noise in the tracking error. This method is the most computationally demanding among those considered, but it achieves a maximum tracking error in all joints of less than 4×10^{-3} radians when disturbances are present, and approximately 7×10^{-5} radians in the undisturbed case. Although the absolute magnitude of these errors is very small (the tiny high-frequency components are barely visible in the reference–following plots), the error signal contains noticeable high-frequency content — short, high-pitch fluctuations that are visible in the error plots but do not materially affect the reference following due to their very small amplitudes.

The control actions produced by the computed-torque law are smooth and, in the absence of disturbances, resemble sinusoidal profiles that closely track the nominal torque/voltage demands of the desired motion. When disturbances are introduced, the dominant sinusoidal component in the control effort tends to diminish and the signals acquire additional variations corresponding to the disturbance rejection behaviour of the controller.

Overall, the computed-torque approach provides the best performance of all methods tested: it yields the lowest steady and transient errors while maintaining smooth control signals. If project resources permit the necessary model accuracy and computational capacity, the computed-torque controller is the preferred choice for high-precision tasks. However, when computational or modelling resources are limited, the PID or nested control solutions still offer very good performance and are viable alternatives.